

COSC 594

Variational Inference (I)

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Outline

- LIKELIHOOD MAXIMIZATION
- EVIDENCE LOWER BOUND

Outline

■ LIKELIHOOD MAXIMIZATION

Generative modeling

Data set



$$x \sim q(\cdot)$$

Generation



$$x \sim p(\cdot)$$

■ DISTRIBUTION DIFFERENCE MINIMIZATION

$$\underset{p(\cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot) \parallel p(\cdot))$$

Difference between two distributions

KL Divergence: $D_{\text{KL}}(q(\cdot) \parallel p(\cdot)) := \mathbb{E}_{x \sim q(\cdot)} \left[\log \frac{q(x)}{p(x)} \right]$

- ★ non-negativeness
- ★ asymmetry
- ★ likelihood maximization

$$D_{\text{KL}}(q(\cdot) \parallel p(\cdot)) = \mathbb{E}_{x \sim q(\cdot)} [\log q(x)] - \underbrace{\mathbb{E}_{x \sim q(\cdot)} [\log p(x)]}_{\text{Log likelihood}}$$

$$\underset{p(\cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot) \parallel p(\cdot)) \quad \Longleftrightarrow \quad \underset{p(\cdot)}{\text{maximize}} \quad \mathbb{E}_{x \sim q(\cdot)} [\log p(x)]$$

Example of LLM pre-training

Token ID	1246	455	848	1734	149	1004	3416	19826	16
Token	It	is	great	to	be	a	Tennessee	Vol	!

$$x \sim q(\cdot)$$

■ AUTOREGRESSIVE MODELS

$$q(x) = q(x_1) q(x_2 | x_1) q(x_3 | x_1, x_2) \cdots q(x_n | x_1, x_2, \dots, x_{n-1})$$

■ LIKELIHOOD MAXIMIZATION

$$\underset{p(\cdot)}{\text{maximize}} \quad \sum_{k=1}^n \mathbb{E}_{x_k \sim q(\cdot | x_1, x_2, \dots, x_{k-1})} [\log p(x_k | x_1, x_2, \dots, x_{k-1})]$$

Latent variable model

Data set



$$(x, \textcolor{red}{z}) \sim q(\cdot, \cdot)$$

Generation



$$(x, \textcolor{red}{z}) \sim p(\cdot, \cdot)$$

$\textcolor{red}{z}$ – latent variable

■ JOINT DISTRIBUTION DIFFERENCE MINIMIZATION

$$\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot))$$

■ JOINT LIKELIHOOD MAXIMIZATION

$$\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \mathbb{E}_{(x, z) \sim q(\cdot, \cdot)} [\log p(x, z)]$$

■ LIKELIHOOD MAXIMIZATION

$$\underset{p(\cdot)}{\text{maximize}} \quad \mathbb{E}_{x \sim q(\cdot)} [\log p(x)]$$

$\Downarrow$

$$\underset{p(\cdot)}{\text{maximize}} \quad \mathbb{E}_{x \sim q(\cdot)} \left[\log \int p(x, z) dz \right] \quad \text{or} \quad \underset{p(\cdot)}{\text{maximize}} \quad \mathbb{E}_{x \sim q(\cdot)} \left[\log \frac{p(x, z)}{p(z | x)} \right]$$

Intractability w/o ‘knowing’ latent variables

Outline

- LIKELIHOOD MAXIMIZATION
- EVIDENCE LOWER BOUND

Evidence (likelihood) lower bound (ELBO)

$$\text{ELBO : } \text{ELBO}(p) := \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q(\cdot | x)} \left[\log \frac{p(x, z)}{q(z | x)} \right]$$

■ LOWER BOUND

$$\begin{aligned} \mathbb{E}_{x \sim q(\cdot)} [\log p(x)] &= \mathbb{E}_{x \sim q(\cdot)} \left[\log \int p(x, z) dz \right] \\ &= \mathbb{E}_{x \sim q(\cdot)} \left[\log \int \frac{p(x, z) q(z | x)}{q(z | x)} dz \right] \\ &= \mathbb{E}_{x \sim q(\cdot)} \left[\log \mathbb{E}_{z \sim q(\cdot | x)} \left[\frac{p(x, z)}{q(z | x)} \right] \right] \\ &\geq \mathbb{E}_{x \sim q(\cdot)} \left[\mathbb{E}_{z \sim q(\cdot | x)} \left[\log \frac{p(x, z)}{q(z | x)} \right] \right] \\ &:= \text{ELBO}(p) \end{aligned}$$

Joint distribution difference $\equiv$ negative ELBO

$$D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot)) = -\text{ELBO}(p) + \mathbb{E}_{x \sim q(\cdot)} [\log q(x)]$$

■ EQUIVALENCE

$$\begin{aligned} D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot)) &= \mathbb{E}_{(x,z) \sim q(\cdot, \cdot)} \left[\log \frac{q(x, z)}{p(x, z)} \right] \\ &= \mathbb{E}_{x \sim q(\cdot), z \sim q(\cdot | x)} \left[\log \frac{q(z | x) q(x)}{p(x, z)} \right] \\ &= \underbrace{\mathbb{E}_{x \sim q(\cdot), z \sim q(\cdot | x)} \left[\log \frac{q(z | x)}{p(x, z)} \right]}_{\text{negative ELBO}} + \mathbb{E}_{x \sim q(\cdot)} [\log q(x)] \end{aligned}$$

$$\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot)) \quad \Longleftrightarrow \quad \underset{p(\cdot, \cdot)}{\text{maximize}} \quad \text{EBLO}(p)$$

Latent gap

Joint KL divergence minimization: $\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot))$

Joint likelihood maximization: $\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \mathbb{E}_{(x, z) \sim q(\cdot, \cdot)} [\log p(x, z)]$

ELBO maximization: $\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \text{ELBO}(p)$

■ LOWER BOUND

$$\underset{p(\cdot)}{\text{maximize}} \quad \mathbb{E}_{x \sim q(\cdot)} [\log p(x)] \quad \geq \quad \underset{p(\cdot, \cdot)}{\text{maximize}} \quad \text{ELBO}(p)$$

Lower bound may have a large gap?

Latent KL divergence

ELBO : $\text{ELBO}(p) := \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q(\cdot | x)} \left[\log \frac{p(x, z)}{q(z | x)} \right]$

■ LATENT GAP

$$\begin{aligned} \mathbb{E}_{x \sim q(\cdot)} [\log p(x)] &= \mathbb{E}_{x \sim q(\cdot)} \left[\int q(z | x) \log p(x) dz \right] \\ &= \mathbb{E}_{x \sim q(\cdot)} \left[\int q(z | x) \log \frac{p(x, z)}{p(z | x)} dz \right] \\ &= \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q(\cdot | x)} \left[\log \frac{p(x, z)}{q(z | x)} + \log \frac{q(z | x)}{p(z | x)} \right] \\ &= \text{ELBO}(p) + \underbrace{\mathbb{E}_{x \sim q(\cdot)} [D_{\text{KL}}(q(\cdot | x) \| p(\cdot | x))]}_{\text{Latent KL divergence}} \end{aligned}$$

Sufficiency of ELBO maximization

Joint KL divergence minimization: $\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot))$

Joint likelihood maximization: $\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \mathbb{E}_{(x, z) \sim q(\cdot, \cdot)} [\log p(x, z)]$

ELBO maximization: $\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \text{ELBO}(p)$

■ CHAIN RULE FOR KL DIVERGENCE

$$\begin{aligned} D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot)) &= \mathbb{E}_{(x, z) \sim q(\cdot, \cdot)} \left[\log \frac{q(x, z)}{p(x, z)} \right] \\ &= \mathbb{E}_{x \sim q(\cdot), z \sim q(\cdot | x)} \left[\log \frac{q(x)q(z | x)}{p(x)p(z | x)} \right] \\ &= D_{\text{KL}}(q(\cdot) \parallel p(\cdot)) + \mathbb{E}_{x \sim q(\cdot)} [D_{\text{KL}}(q(\cdot | x) \parallel p(\cdot | x))] \\ &\geq D_{\text{KL}}(q(\cdot) \parallel p(\cdot)) \end{aligned}$$

Sufficiency of ELBO maximization

Joint KL divergence minimization: $\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot))$

Joint likelihood maximization: $\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \mathbb{E}_{(x, z) \sim q(\cdot, \cdot)} [\log p(x, z)]$

ELBO maximization: $\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \text{ELBO}(p)$

■ CHAIN RULE FOR KL DIVERGENCE

$$\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot)) \quad \Rightarrow \quad \underset{p(\cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot) \parallel p(\cdot))$$



$$\underset{p(\cdot)}{\text{maximize}} \quad \mathbb{E}_{x \sim q(\cdot)} [\log p(x)]$$

Summary

■ LIKELIHOOD MAXIMIZATION

Likelihood maximization $\equiv$ KL divergence minimization, latent variable model

■ EVIDENCE LOWER BOUND

Joint likelihood maximization $\equiv$ joint KL divergence minimization $\equiv$ ELBO maximization, sufficiency of ELBO maximization